

Practice Questions

SS ILP Physics (Intermediate), Week 6 Class 4



Case Problem 1

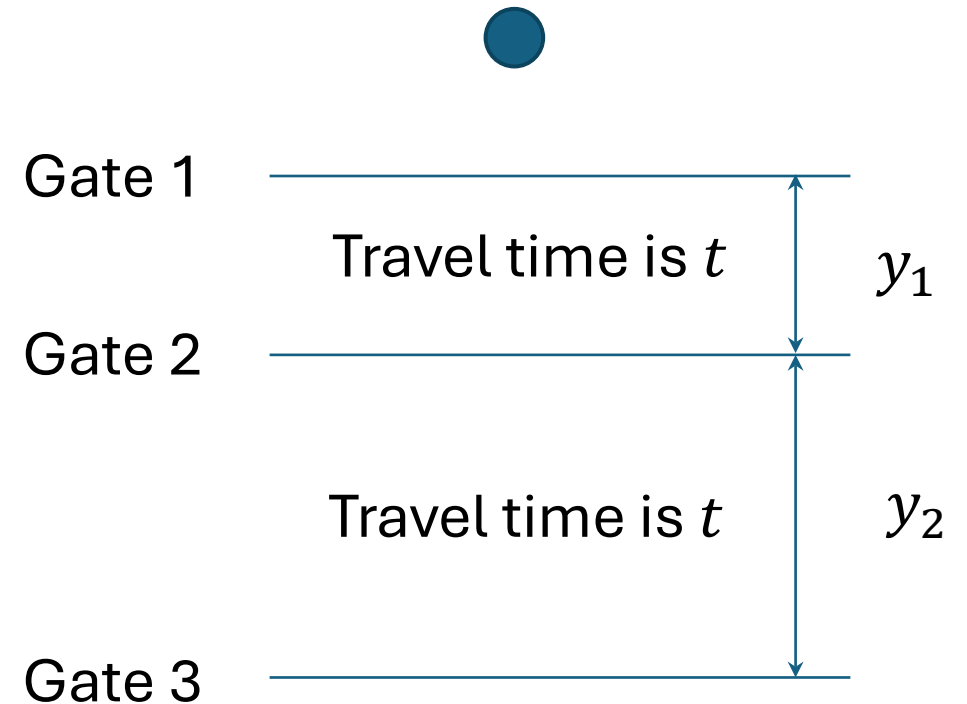
A ball falls down freely in vertical direction at a constant acceleration a . It takes t to fall between gates 1 and 2 that are y_1 apart. It then takes another t to fall between gates 2 and 3 that are y_2 apart. Find the acceleration a as a function of y_1 , y_2 and t .

A. $\frac{y_2 - y_1}{t^2}$

B. $\frac{y_2 - y_1}{2t^2}$

C. $\frac{2(y_2 - y_1)}{3t^2}$

D. $\frac{2(y_2 - y_1)}{t^2}$



<http://a-leveltuition.com/wp-content/uploads/2011/12/2010-A-Level-PH-P1.pdf>

Case Problem 2

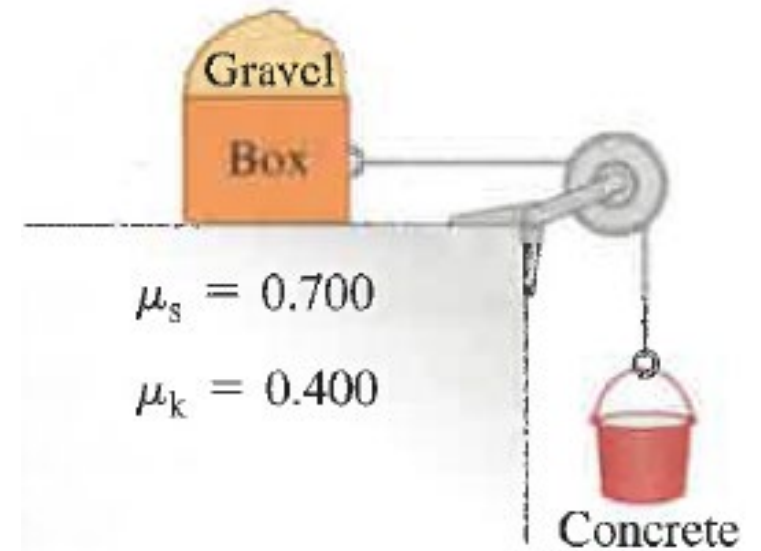
A 20.0-kg crate sits at rest at the bottom of a 15.0-m-long ramp that is inclined at 34 degree above the horizontal. A constant horizontal force of 290 N is applied to the crate to push it up the ramp. While the crate is moving, the ramp exerts a constant frictional force on it that has magnitude 65.0 N. Take $g = 9.8\text{m/s}^2$.

- (a) What is the total work done on the crate during its motion from the bottom to the top of the ramp?
- (b) How much time does it take the crate to travel to the top of the ramp?

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Case Problem 3

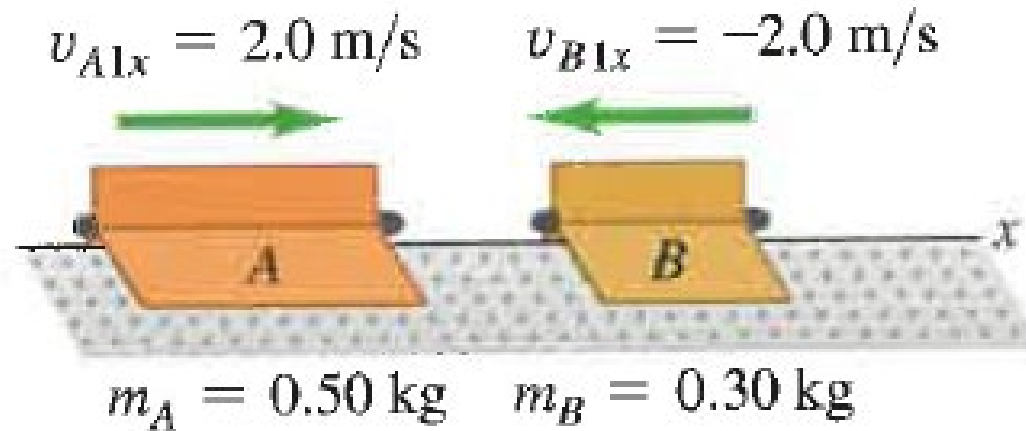
At a construction site, a 67-kg bucket of concrete hangs from a light (but strong) cable that passes over a light, friction-free pulley and is connected to an 82-kg box on a horizontal roof. The cable pulls horizontally on the box, and a 45-kg bag of gravel rests on top of the box. The coefficients of friction between the box and roof are shown. (a) Find the friction force on the bag of gravel and on the box. (b) Suddenly a worker picks up the bag of gravel. Find the speed of the bucket after it has descended 1.6 m from rest. Take $g = 9.8 \text{ m/s}^2$



H. D. Young and R. A. Freedman, University Physics (14th edition): Problem 7.37, p255

Case Problem 4

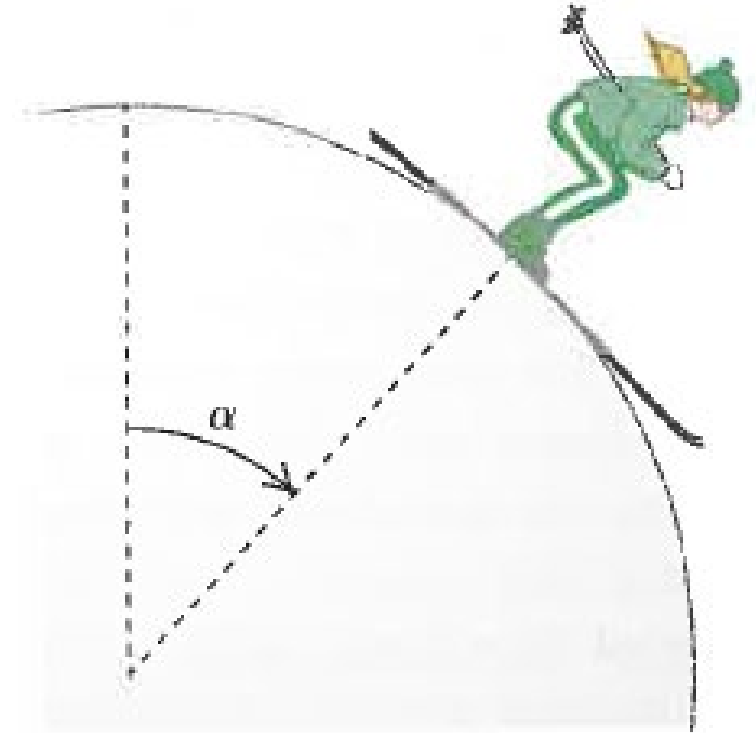
Two gliders with different masses move toward each other on a frictionless air track. Ideal spring bumpers are added to the gliders so that the collision is elastic. What are the final velocities of the gliders?



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 8.10, p277

Case Problem 5

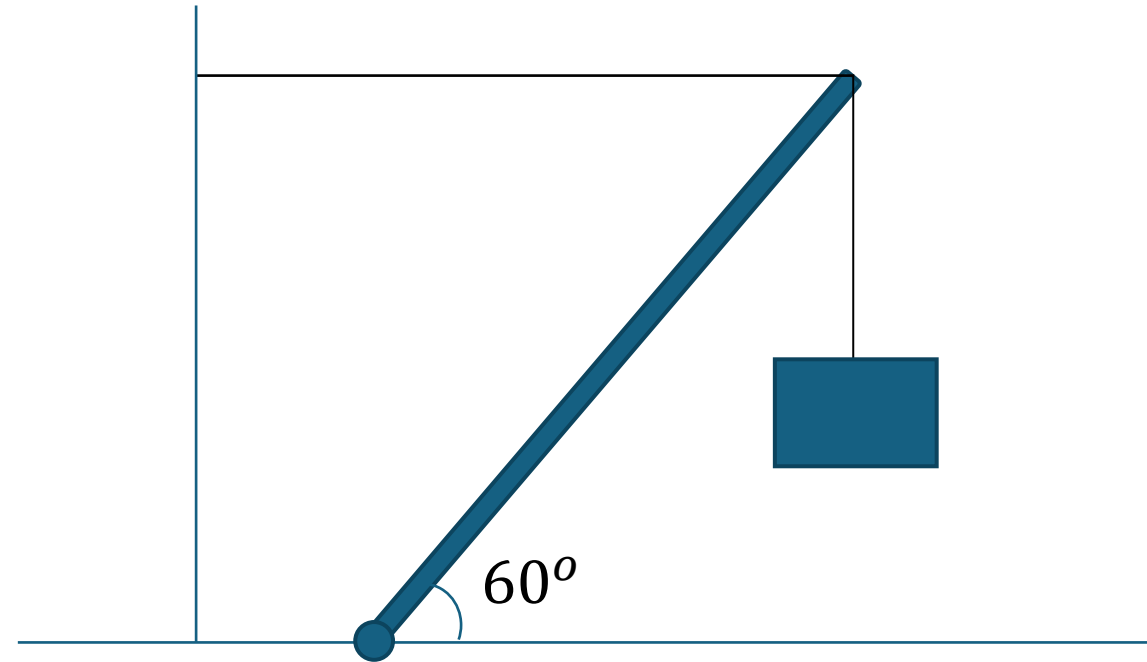
A skier starts at the top of a very large, frictionless snowball, with a very small initial speed, and skis straight down the side. At what point does she lose contact with the snowball and fly off at a tangent? That is, at the instant she loses contact with the snowball, what angle α does a radial line from the center of the snowball to the skier make with the vertical?



H. D. Young and R. A. Freedman, University Physics (14th edition): Problem 7.55, p256

Case Problem 6

A system of a rod (of length l and mass m_1) and a box (of mass m_2) is in static equilibrium, as shown. Find the tension due to the horizontal string and the force due to the hinge.



<https://www2.tntech.edu/leap/murdock/books/v2chap3.pdf>

Off-class practice question(s)

Case Problem

A factory worker pushes a 30.0-kg crate a distance of 4.5 m along a level floor at constant velocity by pushing horizontally on it. The coefficient of kinetic friction between the crate and the floor is 0.25. Take $g = 9.8\text{m/s}^2$.

- (a) What magnitude of force must the worker apply?
- (b) How much work is done on the crate by this force?
- (c) How much work is done on the crate by friction?
- (d) How much work is done on the crate by the normal force? By gravity?
- (e) What is the total work done on the crate?

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Case Problem: Solution

- (a) The factory worker must at least exert a force equals to the friction, $f_k = \mu_k N$, where $N = mg$ is the normal force. The magnitude of the force the worker must apply is hence

$$F_{worker} = f_k = \mu_k mg = 0.25 \cdot 30.0 \text{ kg} \cdot 9.80 \text{ m/s}^2 = 74 \text{ N}$$

- (b) Since the force applied by the worker is horizontal and in the direction of the displacement, $\phi = 0^\circ$ and the work is:

$$W_{worker} = F_{worker} s \cos 0^\circ = 74 \text{ N} \cdot 4.5 \text{ m} \cdot 1 = 333 \text{ J}$$

- (c) Friction acts in the direction opposite of motion, thus $\phi = 180^\circ$ and the work of friction is:

$$W_{friction} = f_k s \cos 180^\circ = 74 \text{ N} \cdot 4.5 \text{ m} \cdot (-1) = -333 \text{ J}$$

Case Problem: Solution

(d) Both gravity and the normal force act perpendicular to the direction of displacement. Thus, neither force does any work on the crate and $W_{grav} = W_N = 0 \text{ J}$.

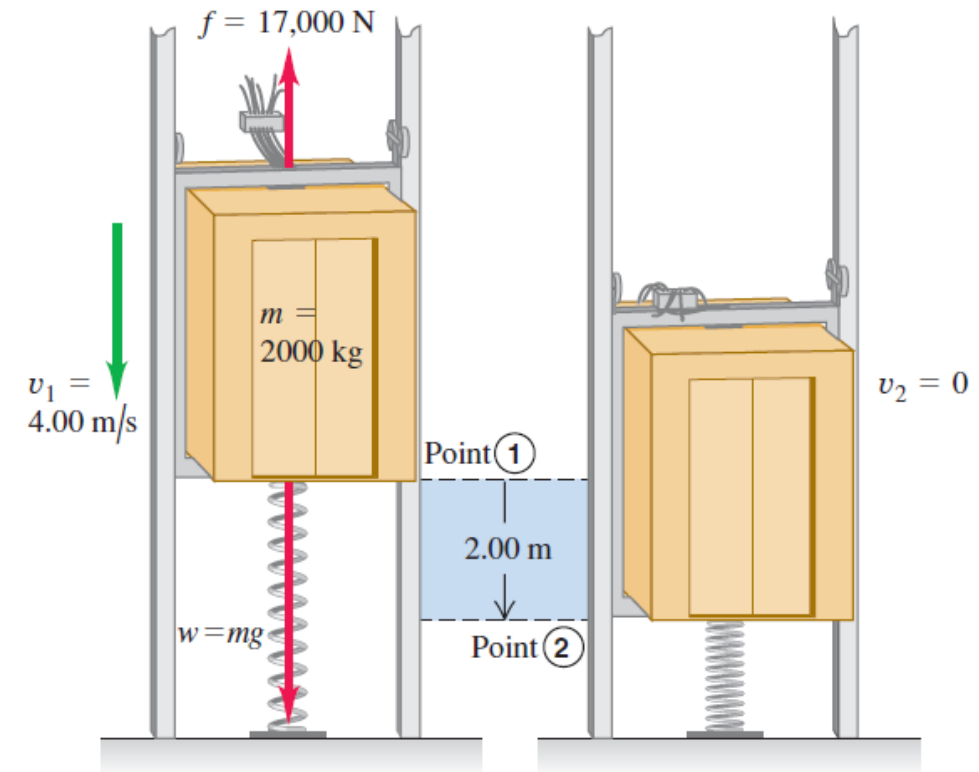
(e) The net work done is

$$W_{total} = W_{worker} + W_{friction} + W_{grav} + W_N = 0 \text{ J}$$

The net work done is zero since there is no net force acting on the crate, i.e. the force by the worker exactly cancels the friction force.

Case Problem

A 2000-kg elevator with broken cables in a test rig is falling at 4.0 m/s when it contacts a cushioning spring at the bottom of the shaft. The spring is intended to stop the elevator, compressing 2.00 m as it does so. During the motion a safety clamp applies a constant 17,000-N frictional force to the elevator. What is the necessary force constant k for the spring?



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Case Problem: Solution

The kinetic energy and the gravitational and elastic potential energies at point 1 and point 2 are (set $y_1 = 0$, $y_2 = -2.00 \text{ m}$):

$$K_1 = \frac{1}{2}mv_1^2$$

$$U_{grav,1} = mgy_1 = 0$$

$$U_{el,1} = \frac{1}{2}ky_1^2 = 0$$

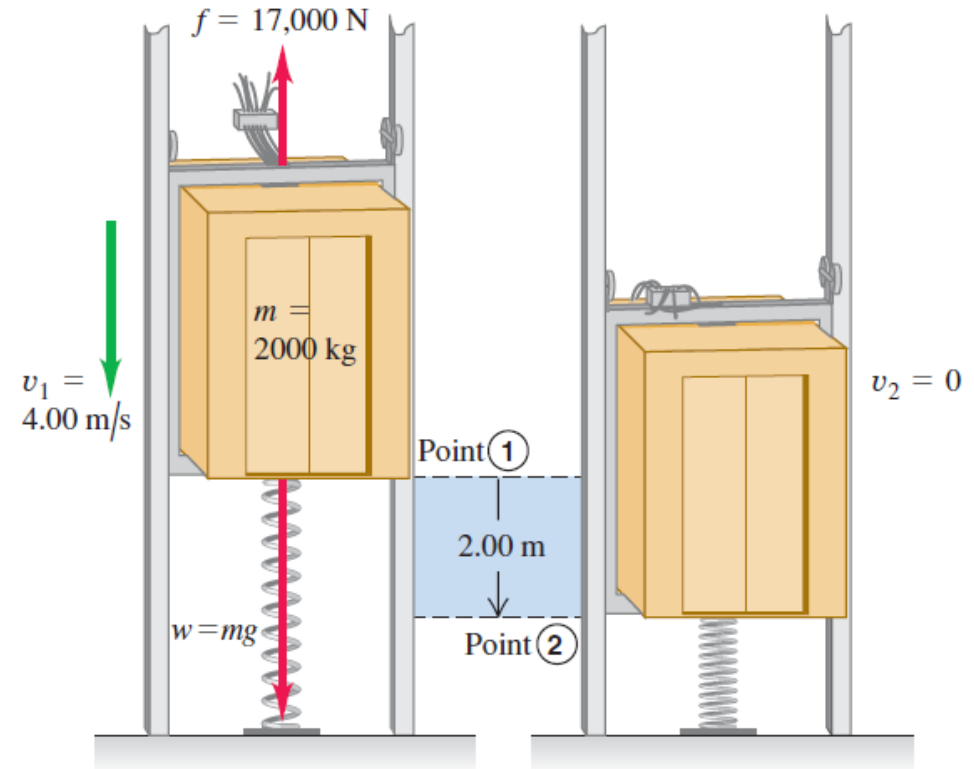
$$K_2 = \frac{1}{2}mv_2^2 = 0$$

$$U_{grav,2} = mgy_2$$

$$U_{el,2} = \frac{1}{2}ky_2^2$$

As the elevator is sliding down, the friction force produces a negative work done,

$$W_f = -f(y_1 - y_2)$$



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Case Problem: Solution

By conservation of energy,

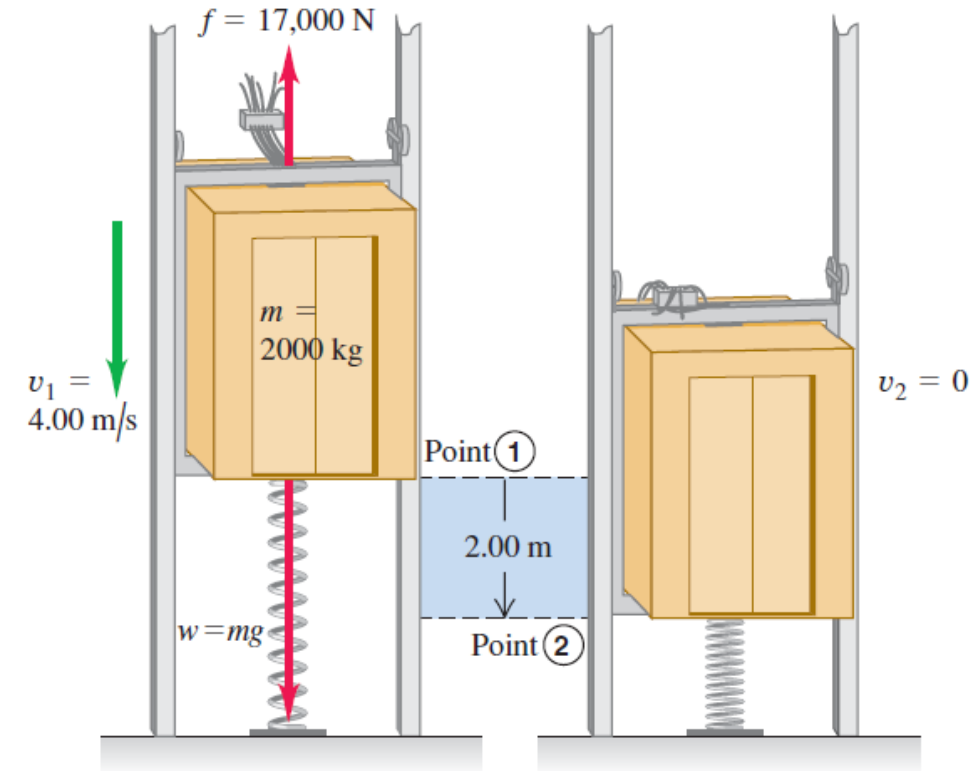
$$K_1 + U_{grav,1} + U_{el,1} + W_f = K_2 + U_{grav,2} + U_{el,2}$$

$$\frac{1}{2}mv_1^2 + 0 + 0 - f(y_1 - y_2) = 0 + mgy_2 + \frac{1}{2}ky_2^2$$

$$\frac{1}{2}mv_1^2 - f(y_1 - y_2) = mgy_2 + \frac{1}{2}ky_2^2$$

$$k = \frac{mv_1^2 - 2f(y_1 - y_2) - 2mgy_2}{y_2^2}$$

$$= 1.06 \times 10^4 \text{ Nm}^{-1}$$



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